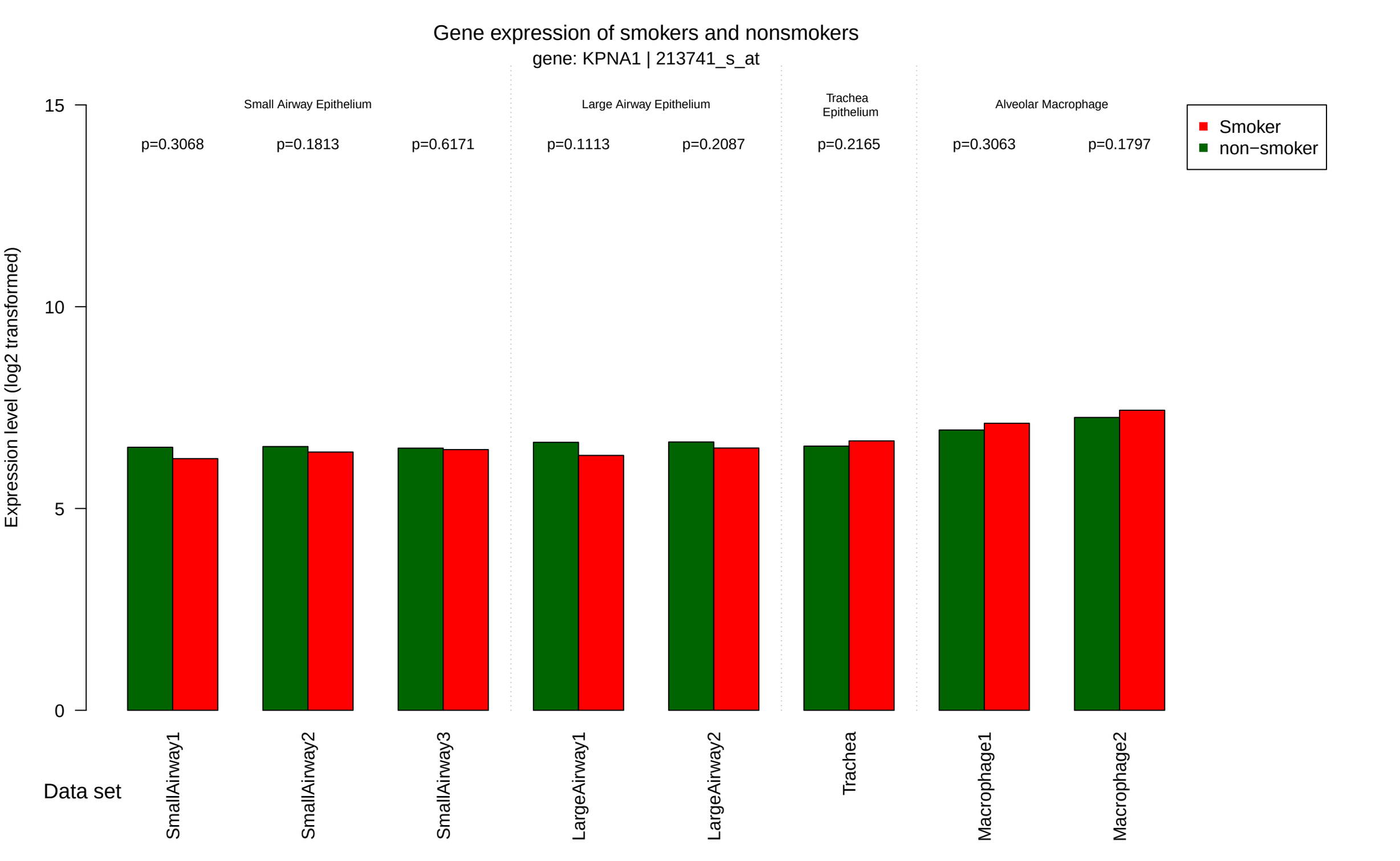
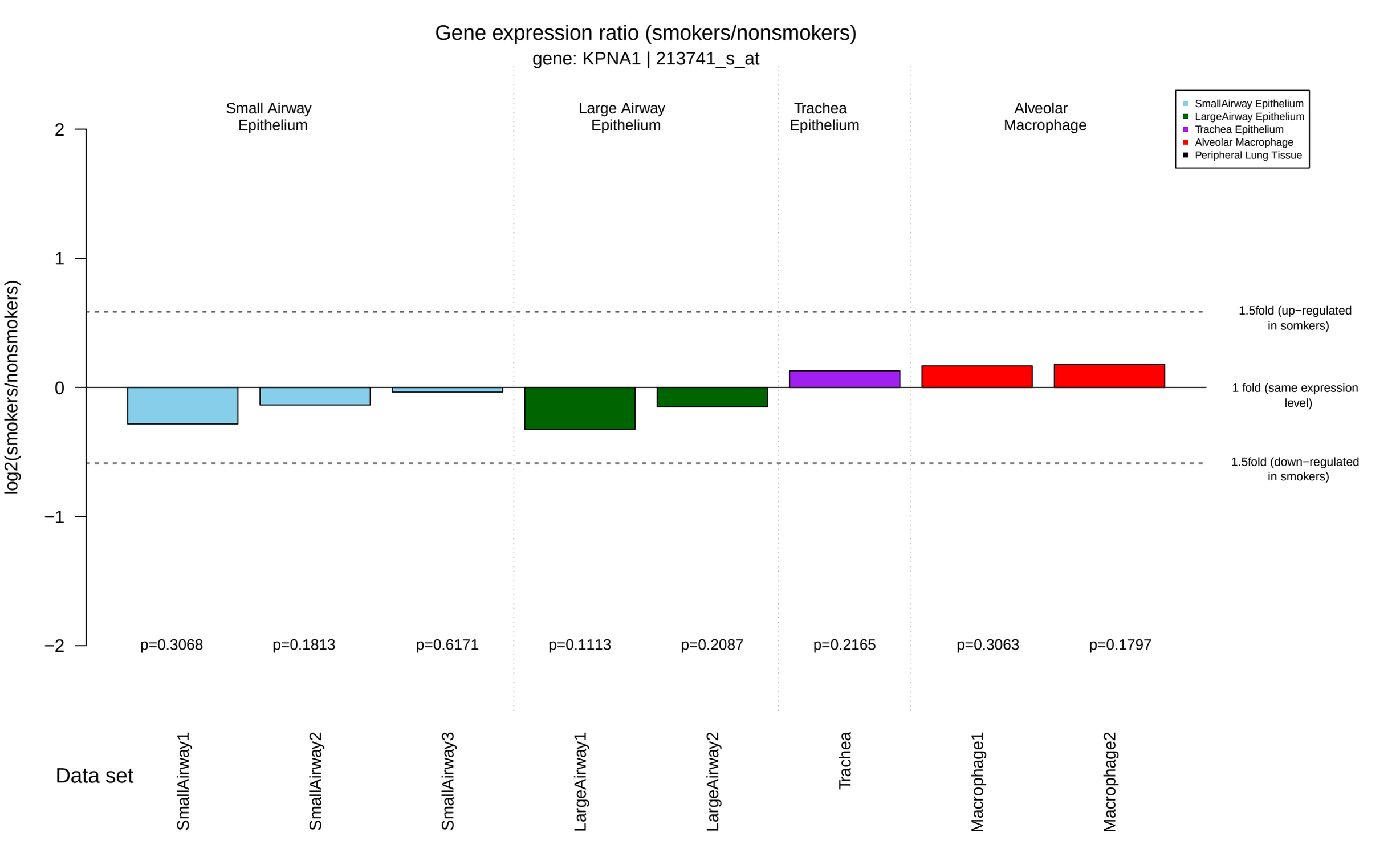
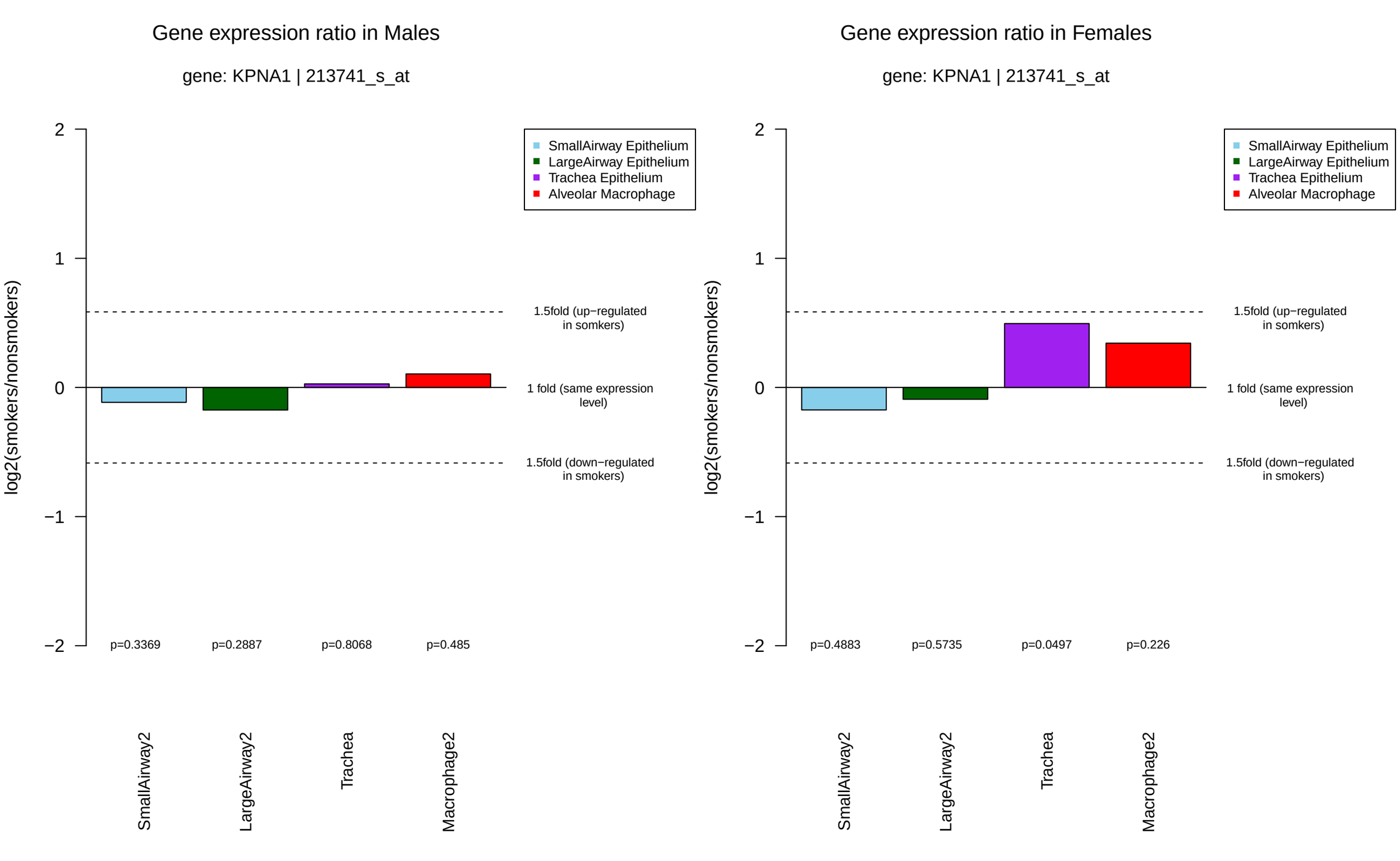
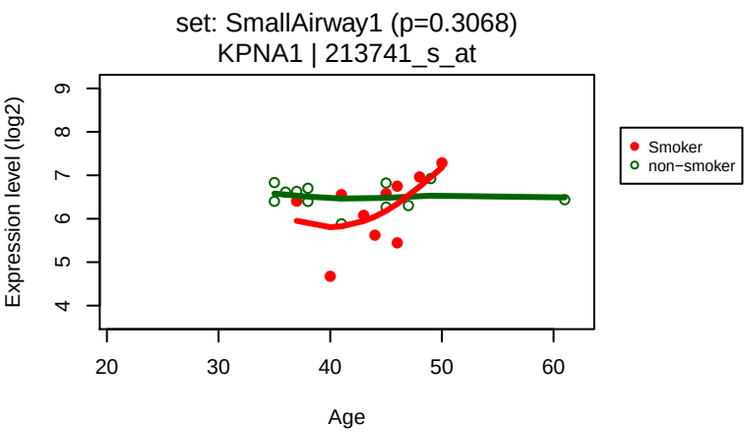
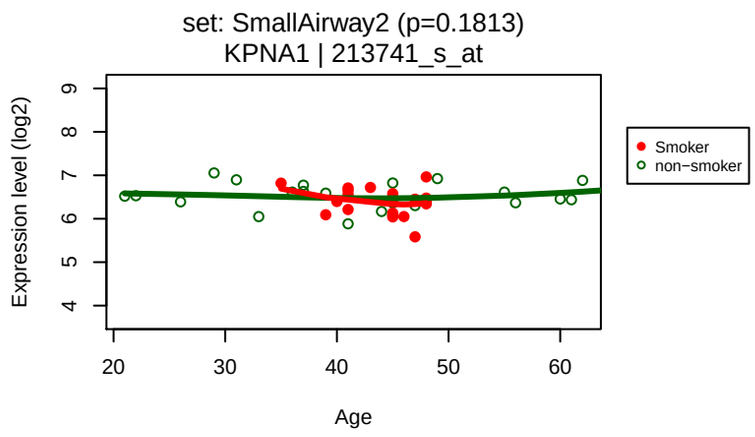
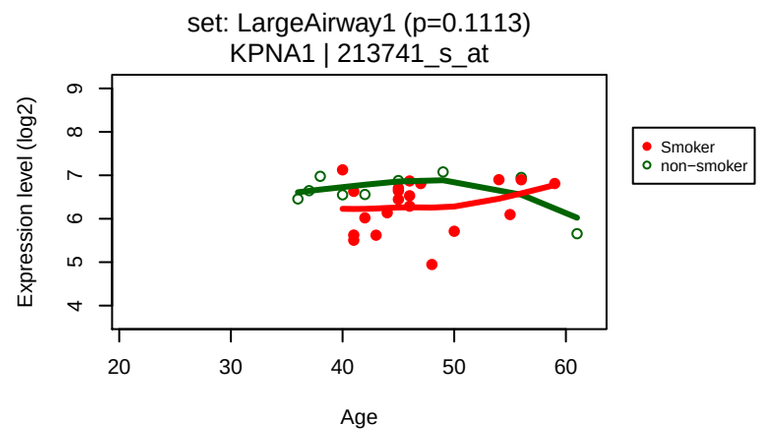

Gene expression of smokers and nonsmokers
gene: KPNA1 | 213741_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3068
p=0.1813
p=0.6171
p=0.1113
p=0.2087
p=0.2165
p=0.3063
p=0.1797
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

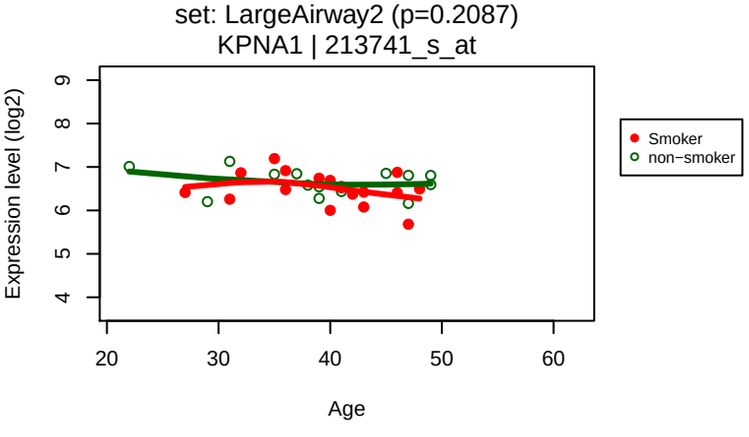

Gene expression ratio (smokers/nonsmokers)
gene: KPNA1 | 213741_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3068
p=0.1813
p=0.6171
p=0.1113
p=0.2087
p=0.2165
p=0.3063
p=0.1797
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

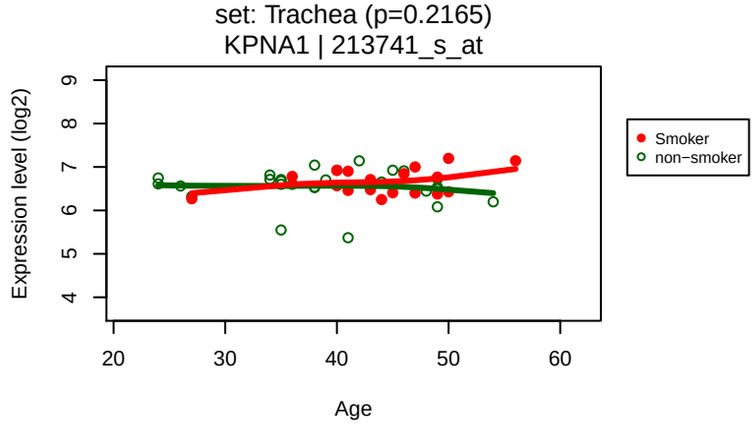

Gene expression ratio in Males
gene: KPNA1 | 213741_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3369
p=0.2887
p=0.8068
p=0.485
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KPNA1 | 213741_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4883
p=0.5735
p=0.0497
p=0.226
SmallAirway2
LargeAirway2
Trachea
Macrophage2

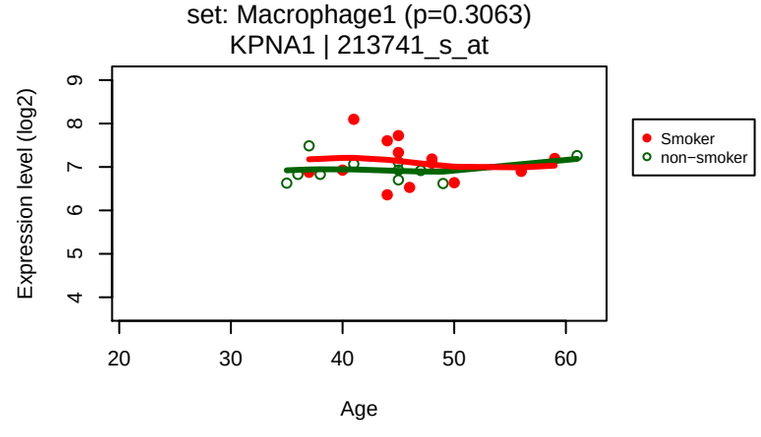

set: SmallAirway1 (p=0.3068)
KPNA1 | 213741_s_at
Expression level (log2)
Age
Smoker
non−smoker

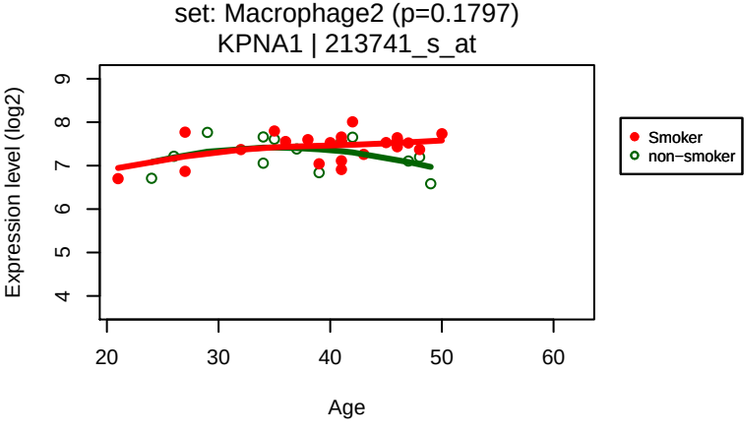

set: SmallAirway2 (p=0.1813)
KPNA1 | 213741_s_at
Expression level (log2)
Age
Smoker
non−smoker

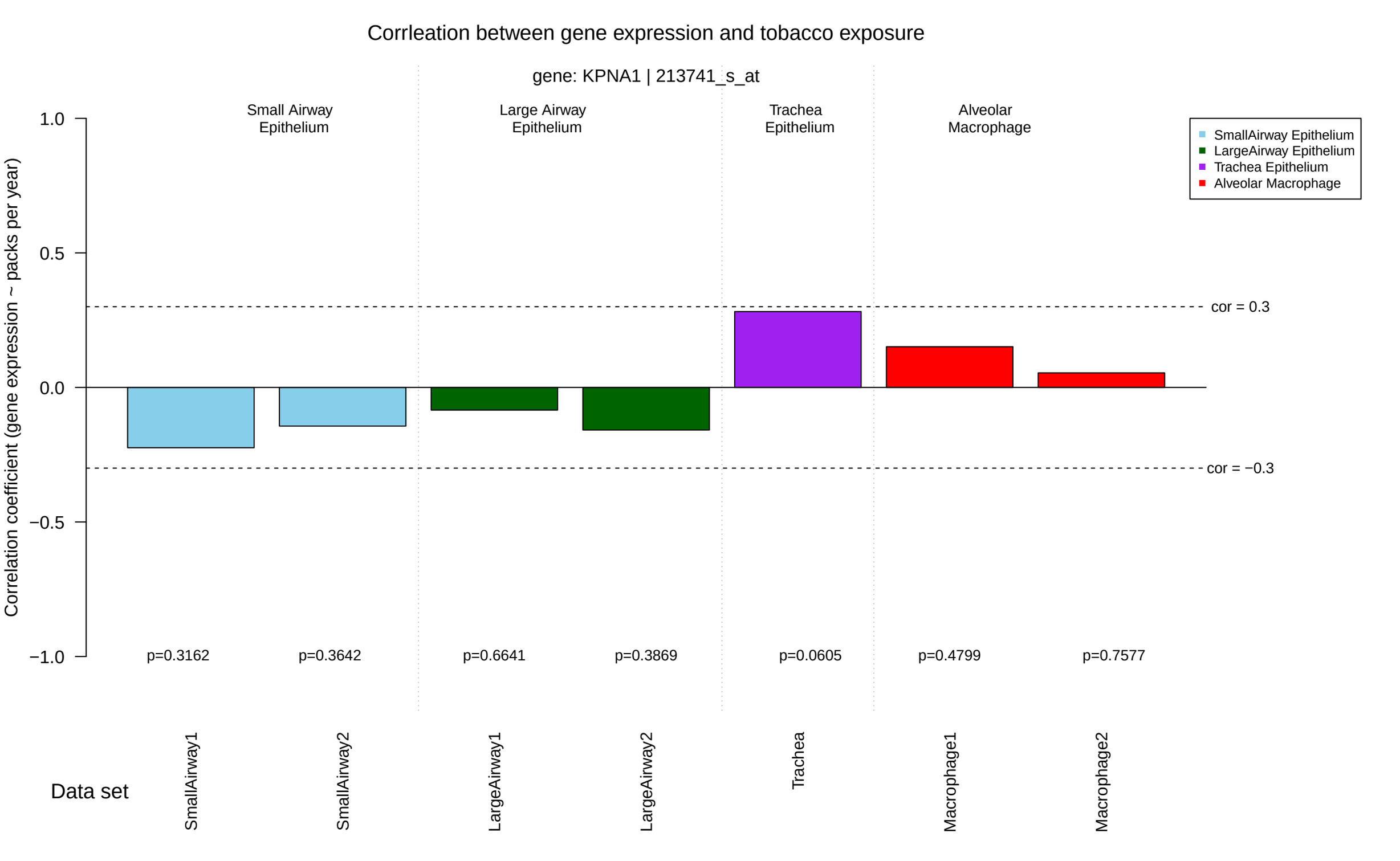

set: LargeAirway1 (p=0.1113)
KPNA1 | 213741_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2087)
KPNA1 | 213741_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2165)
KPNA1 | 213741_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3063)
KPNA1 | 213741_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1797)
KPNA1 | 213741_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KPNA1 | 213741_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3162
p=0.3642
p=0.6641
p=0.3869
p=0.0605
p=0.4799
p=0.7577
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2